

Task 1 – Project Workspace

6 Marks

Create:

<FirstName>-GLA-LibrarySystem

Inside it create:

- src
- docs
- config
- backup

Create:

library_info.txt

Include:

- Name
- Roll Number
- Subject
- Project Name
- Semester

Demonstrate Linux commands for:

- pwd
- directory navigation
- listing files
- displaying file contents
- displaying directory structure

SOLUTION:

```
Last login: Mon Sep  7 18:38:46 on ttys002
paridhishukla@Paridhi-MacBook-Air ~ % mkdir Paridhi-GLA-LibrarySystem
paridhishukla@Paridhi-MacBook-Air ~ % cd Paridhi-GLA-LibrarySystem
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % mkdir src docs config backup
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % touch library_info.txt
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % ls
backup      docs        library_info.txt  src
config
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % nano library_info.txt
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % cat library_info.txt
Name: Paridhi Shukla
Roll Number: YOUR_ROLL_NUMBER
Subject: Application Build and Release Automation Lab
Project Name: Online Library Management System
Semester: 6th
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % pwd
/Users/paridhishukla/Paridhi-GLA-LibrarySystem
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % cd src
paridhishukla@Paridhi-MacBook-Air src % pwd
/Users/paridhishukla/Paridhi-GLA-LibrarySystem/src
paridhishukla@Paridhi-MacBook-Air src % cd ..
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % pwd
/Users/paridhishukla/Paridhi-GLA-LibrarySystem
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % ls
backup      docs        library_info.txt  src
config
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % ls -l
total 8
drwxr-xr-x  2 paridhishukla  staff   64 Sep 11 11:57 backup
drwxr-xr-x  2 paridhishukla  staff   64 Sep 11 11:57 config
drwxr-xr-x  2 paridhishukla  staff   64 Sep 11 11:57 docs
-rw-r--r--  1 paridhishukla  staff  166 Sep 11 12:00 library_info.txt
drwxr-xr-x  2 paridhishukla  staff   64 Sep 11 11:57 src
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % tree
.
├── backup
├── config
├── docs
├── library_info.txt
└── src

5 directories, 1 file
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem %
```

Task 2 – Git Repository

6 Marks

1. Initialize Git.
2. Configure Git username/email.
3. Check status.
4. Stage project files.
5. Create the initial commit.
6. Modify `library_info.txt`.
7. Add a project objective and description.
8. Check the changes.
9. Stage the modified file.
10. Create a second commit.
11. Display commit history.

SOLUTION:

```
$ directories, 1 file
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to 'main' in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /Users/paridhishukla/Paridhi-GLA-LibrarySystem/.git/
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % git branch -m main
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % git branch
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % git config --global user.name "Paridhi Shukla"
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % git config --global user.email "YOUR_EMAIL"
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % git config --global user.name "shuklaparidhi2008@gmail.com"
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % git config --global user.name
Paridhi Shukla
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % git config --global user.email
shuklaparidhi2008@gmail.com
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    library_info.txt

nothing added to commit but untracked files present (use "git add" to track)
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % git add .
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   library_info.txt

paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % git commit -m "Initial project setup"
[master (root-commit) 90758dc] Initial project setup
 1 file changed, 5 insertions(+)
 create mode 100644 library_info.txt
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % git status
On branch main
nothing to commit, working tree clean
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % nano library_info.txt
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % cat library_info.txt
Name: Paridhi Shukla
Roll Number: YOUR_ROLL_NUMBER
Subject: Application Build and Release Automation Lab
Project Name: Online Library Management System
Semester: 5th
Project Objective:
To develop an Online Library Management System for efficiently managing library resources and book-related information.

Project Description:
The Online Library Management System is designed to manage books and provide an organized platform for library operations.
paridhishukla@Paridhi-MacBook-Air Paridhi-GLA-LibrarySystem % git status
```

```

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   library_info.txt

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git commit -m "Initial project setup"
[main (root-commit) 907358c] Initial project setup
 2 file changed, 5 insertions(+)
 create mode 100644 library_info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git status
On branch main
nothing to commit, working tree clean
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % nano library_info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % cat library_info.txt
Name: Paridhi Shukla
Roll Number: YOUR_ROLL_NUMBER
Subject: Application Build and Release Automation Lab
Project Name: Online Library Management System
Semester: 5th
Project Objective:
To develop an Online Library Management System for efficiently managing library resources and book-related information.

Project Description:
The Online Library Management System is designed to manage books and provide an organized platform for library operations.
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   library_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git diff
diff --git a/library_info.txt b/library_info.txt
index 0b72b82..92a17ae 100644
--- a/library_info.txt
+++ b/library_info.txt
@@ -3,3 +3,8 @@ Roll Number: YOUR_ROLL_NUMBER
 Subject: Application Build and Release Automation Lab
 Project Name: Online Library Management System
 Semester: 5th
+Project Objective:
+To develop an Online Library Management System for efficiently managing library resources and book-related information.
+
+Project Description:
+The Online Library Management System is designed to manage books and provide an organized platform for library operations.
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git add library_info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   library_info.txt

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git commit -m "Added project objective and description"
[main c01a765] Added project objective and description
 1 file changed, 5 insertions(+)
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git status
On branch main
nothing to commit, working tree clean
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git log --oneline
c01a765 (HEAD -> main) Added project objective and description
907358c Initial project setup
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % █

```

Task 3 – Feature Branch

6 Marks

Develop a **Book Management Module**.

Create branch:

book-feature

Inside **src**, create:

book.txt

The file must describe the Book Management Module and contain at least three features.

Stage and commit the changes.

Demonstrate:

- branch creation
- branch switching
- current branch
- all branches
- returning to main

SOLUTION:

```

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git commit -m "Added project objective and description"
[main c61a768] Added project objective and description
1 file changed, 2 insertions(+)
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git status
On branch main
nothing to commit, working tree clean
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git log --oneline
c61a768 (HEAD -> main) Added project objective and description
987358c Initial project setup
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git branch book-feature
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git branch
* book-feature
# main
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git switch book-feature
Switched to branch 'book-feature'
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git branch --show-current
book-feature
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % touch src/book.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % ls src
book.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % nano src/book.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % cat src/book.txt
Book Management Module

Description:
The Book Management Module is a component of the Online Library Management System used to manage and organize book-related information.

Features:
1. Add New Book
   - Allows the librarian to add new books to the library system.
2. Update Book Information
   - Allows the librarian to modify existing book details.
3. Delete Book
   - Allows the librarian to remove a book from the system.
4. Search Book
   - Allows users to search for books using relevant book information.
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git status
On branch book-feature
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    src/

nothing added to commit but untracked files present (use "git add" to track)
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git add src/book.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git status
On branch book-feature
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   src/book.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git commit -m "Added Book Management Module"
[book-feature 89dae5b] Added Book Management Module
1 file changed, 27 insertions(+)
create mode 100644 src/book.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git branch --show-current
book-feature
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git branch
* book-feature
# main
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git switch main
Switched to branch 'main'
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % █

```

Task 4 – Merge Conflict

8 Marks

1. Merge **book-feature** into main.
2. Create:

library-update

3. Modify the same line in **library_info.txt** in **library-update**.
4. Commit the change.
5. Modify that same line differently in main.
6. Commit the change.
7. Attempt the merge.
8. Identify the conflict.
9. Resolve it manually.
10. Stage and commit the resolved file.
11. Display the final file.
12. Display Git status and history.

SOLUTION:

```

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git switch main
Switched to branch 'main'
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git branch --show-current
main
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % nano library_info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % cat library_info.txt
Name: Paridhi Shukla
Roll Number: YOUR_ROLL_NUMBER
Subject: Application Build and Release Automation Lab
Project Name: Digital Library Management System
Semester: 5th
Project Objective:
To develop an Online Library Management System for efficiently managing library resources and book-related information.

Project Description:
The Online Library Management System is designed to manage books and provide an organized platform for library operations.
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git commit -m "Updated project name in main"
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   library_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git add library_info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git add library_info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   library_info.txt

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git commit -m "Updated project name in main"
[main 1918758] Updated project name in main
 1 file changed, 1 insertion(+), 1 deletion(-)
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git merge library-update
Auto-merging library_info.txt
CONFLICT (content): Merge conflict in library_info.txt
Automatic merge failed; fix conflicts and then commit the result.
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git merge library-update
error: Merging is not possible because you have unmerged files.
hint: fix them up in the work tree, and then use 'git add/rm <file>'
hint: as appropriate to mark resolution and make a commit.
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git status
On branch main
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   library_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % cat library_info.txt
Name: Paridhi Shukla
Roll Number: YOUR_ROLL_NUMBER
Subject: Application Build and Release Automation Lab
<<<<< HEAD
Project Name: Digital Library Management System
=====
Project Name: Advanced Library Management System
>>>>> library-update
Semester: 5th
Project Objective:
To develop an Online Library Management System for efficiently managing library resources and book-related information.

Project Description:
The Online Library Management System is designed to manage books and provide an organized platform for library operations.
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % nano library_info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % cat library_info.txt
Name: Paridhi Shukla
Roll Number: YOUR_ROLL_NUMBER
Subject: Application Build and Release Automation Lab
Project Name: Digital Library Management System
Project Name: Advanced Library Management System
library-update
Semester: 5th
Project Objective:
To develop an Online Library Management System for efficiently managing library resources and book-related information.

Project Description:
The Online Library Management System is designed to manage books and provide an organized platform for library operations.
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git add library_info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git merge library-update
All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

Changes to be committed:
  modified:   library_info.txt

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git commit -m "Resolved merge conflict"
[main e4eadd] Resolved merge conflict
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git status
On branch main
nothing to commit, working tree clean
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git log --oneline --graph --all
* e4eadd (HEAD -> main) Resolved merge conflict
|
| * f0cb516 (library-update) Updated project name in library-update
| * 1918758 Updated project name in main
| /
* 89daeb8 (book-feature) Added Book Management Module
* c61a765 Added project objective and description
* 987550c Initial project setup
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem %

```

```

Subject: Application Build and Release Automation Lab
<<<<< HEAD
Project Name: Digital Library Management System
=====
Project Name: Advanced Library Management System
>>>>> library-update
Semester: 5th
Project Objective:
To develop an Online Library Management System for efficiently managing library resources and book-related information.

Project Description:
The Online Library Management System is designed to manage books and provide an organized platform for library operations.
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % nano library_info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % cat library_info.txt
Name: Paridhi Shukla
Roll Number: YOUR_ROLL_NUMBER
Subject: Application Build and Release Automation Lab
Project Name: Digital Library Management System
Project Name: Advanced Library Management System
library-update
Semester: 5th
Project Objective:
To develop an Online Library Management System for efficiently managing library resources and book-related information.

Project Description:
The Online Library Management System is designed to manage books and provide an organized platform for library operations.
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git add library_info.txt
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git merge library-update
All conflicts fixed but you are still merging.
  (use "git commit" to conclude merge)

Changes to be committed:
  modified:   library_info.txt

paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git commit -m "Resolved merge conflict"
[main e4eadd] Resolved merge conflict
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git status
On branch main
nothing to commit, working tree clean
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem % git log --oneline --graph --all
* e4eadd (HEAD -> main) Resolved merge conflict
|
| * f0cb516 (library-update) Updated project name in library-update
| * 1918758 Updated project name in main
| /
* 89daeb8 (book-feature) Added Book Management Module
* c61a765 Added project objective and description
* 987550c Initial project setup
paridhishukla@Paridhis-MacBook-Air Paridhi-GLA-LibrarySystem %

```

Task 5 – Submission

4 Marks

Submit:

- Screenshots of important commands.
- Project directory.
- Git history.
- Branch information.
- Merge demonstration.
- Conflict and resolution demonstration.

Students must be able to explain the commands used.

Total: 30 Marks

SOLUTION: